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**To cite this article:** Yoshimasa Uematsu & Takashi Yamagata (2025) Discovering the Network Granger Causality in Large Vector Autoregressive Models, Journal of the American Statistical Association, 120:552, 2385-2396, DOI: [10.1080/01621459.2025.2450836](https://doi.org/10.1080/01621459.2025.2450836)

**To link to this article:** <https://doi.org/10.1080/01621459.2025.2450836>



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# Discovering the Network Granger Causality in Large Vector Autoregressive Models

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## ABSTRACT

This article proposes novel inferential procedures for discovering the network Granger causality in high-dimensional vector autoregressive models. In particular, we mainly offer two multiple testing procedures designed to control the false discovery rate (FDR). The first procedure is based on the limiting normal distribution of the  $t$ -statistics with the debiased lasso estimator. The second procedure is its bootstrap version. We also provide a robustification of the first procedure against any cross-sectional dependence using asymptotic  $e$ -variables. Their theoretical properties, including FDR control and power guarantee, are investigated. The finite sample evidence suggests that both procedures can successfully control the FDR while maintaining high power. Finally, the proposed methods are applied to discovering the network Granger causality in a large number of macroeconomic variables and regional house prices in the United Kingdom. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

## ARTICLE HISTORY

Received September 2022  
Accepted January 2025

## KEYWORDS

Bootstrap; Debiased lasso;  
E-values; FDR and power;  
Multiple testing

## 1. Introduction

Revealing a dynamic interrelationship among variables is a critical challenge in economics, finance, neuroscience, genomics, and so forth. In particular, identifying the ability of a time series variable to predict the future values of other time series variables has been of great interest in various fields. Following the vast literature after Granger (1969), when the past values of a time series  $x := \{x_t\}$  can predict the future values of another time series  $y$ , this will be expressed as “ $x$  is *Granger-causal* for  $y$ ” in this article. This Granger causality has conventionally been discussed in a bivariate relationship, typically with (bivariate) vector autoregressive (VAR) models; see Sims (1972) and Hosoya (1977), among many others.

### 1.1. Network Granger Causality

Suppose that an  $N$ -dimensional time series  $\mathbf{y}$  follows the stationary VAR( $K$ ) model:

$$\mathbf{y}_t = \Phi_1 \mathbf{y}_{t-1} + \cdots + \Phi_K \mathbf{y}_{t-K} + \mathbf{u}_t,$$

where  $\Phi_k = (\phi_{ij,k})$  is the  $k$ th coefficient matrix and  $\mathbf{u}_t$  is the error vector. The predictability relationship among the  $N$  time series manifests as the sparsity pattern of the coefficient matrices in the VAR( $K$ ) model. The *Granger-causal network* is defined as the graph  $G = (V, E)$  with vertex set  $V = [N] := \{1, \dots, N\}$  and edge set  $E$  such that for distinct  $i, j \in V$ ,  $i \rightarrow j \in E$  if and only if  $\phi_{ji,k} \neq 0$  for some  $k \in [K]$ . The same definition has been adopted by Basu, Shojaie, and Michailidis (2015) and Eichler

(2007, 2012b), for instance. See also Shojaie and Fox (2021) for recent advances in the analysis of network Granger causality. As an illustration, we consider the four-dimensional VAR(2) model with the sparsity pattern of the coefficient matrices shown in Figure 1(a), where the nonzero and zero elements are indicated as the gray and white cells, respectively. From this sparse structure, we readily obtain the associated Granger-causal network in Figure 1(b).

Although the network Granger causality has been simply defined as the *direct* causal network as above, we have to remark the existence of further *indirect* causalities over the lagged variables. For the aforementioned VAR(2) model, Figure 2 depicts the causal chain that takes all the lags into account (Eichler 2012a, sec. 3), where the chain continues to the infinite past and future due to the stationarity. In this figure, we can identify many indirect causalities; for instance,  $y_{3,t-2}$  is indirectly causal to  $y_{2t}$  via  $y_{1,t-1}$ . To refine the concept of such direct and indirect causalities, Dufour and Renault (1998) introduced short- and long-run causality called the  $h$ -step ( $h \geq 1$ ) causality. In this context, the  $h$ -step non-causality is characterized by zero restrictions to a part of the first  $N$  rows of the first to  $h$ th powers of the companion coefficient matrices (Lütkepohl 2005, p.50). Indeed, in Figure 2, it can be shown that the one-step (i.e., direct) causality is expressed by the thick arrows while the indirect effects will be included in  $h$ -step ( $h \geq 2$ ) causalities. Apparently, this one-step causality corresponds to the network Granger causality we have defined.

The Granger-causal network can be discovered just by estimating the sparse  $\Phi_k$ 's in an appropriate manner. In early empir-

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 Supplementary materials for this article are available online. Please go to [www.tandfonline.com/r/JASA](http://www.tandfonline.com/r/JASA).

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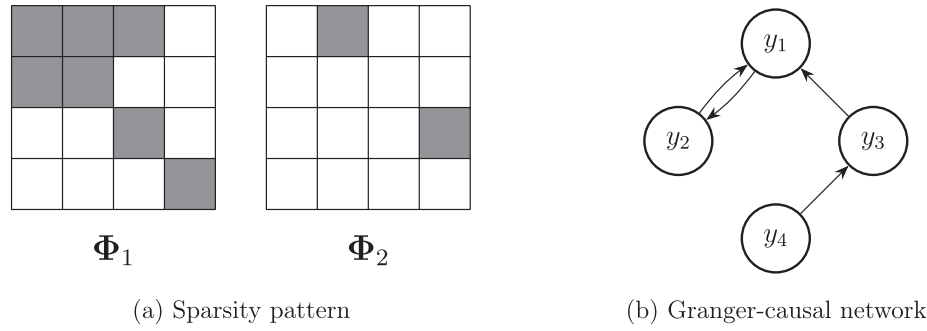


Figure 1. Sparse VAR(2) coefficients and associated Granger-causal network.

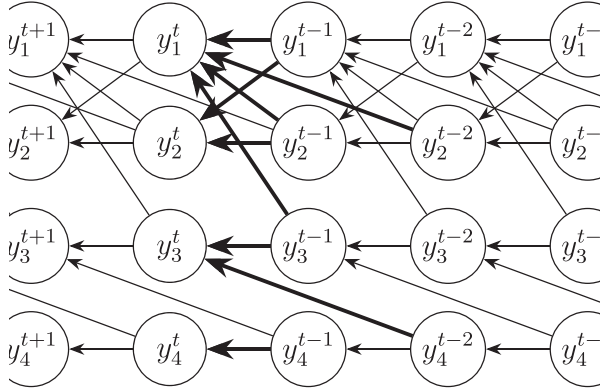


Figure 2. Causal chain of the VAR(2).

ical studies, including Fujita et al. (2007) and Lozano et al. (2009) among others, such estimation methods were already employed and a network was estimated. Recently the theoretical properties of these methods have begun to be investigated. Basu and Michailidis (2015), Kock and Callot (2015), and Basu, Li, and Michailidis (2019) study properties of the lasso (Tibshirani 1996) and its variants in high-dimensional VAR models. Han, Lu, and Liu (2015) extends the Dantzig selector of Candès and Tao (2007) to be applicable to (weakly) sparse VAR models. Kock and Callot (2015) and Barigozzi and Brownlees (2019) propose the adaptive lasso estimator. Davis, Zang, and Zheng (2016) consider an alternative two-step estimator, which uses estimates of the partial spectral coherence.

### 1.2. Classical Granger Causality Test

Estimation-based network detection, such as the lasso selection, is appealing because of its simplicity. However, along with the difficulty of evaluating the type I error of the selection result, this may cause instability and a lack of reproducibility. Thus, we discuss the inference-based network detection. Indeed, statistical inference of Granger causality was the central issue, as Sir Clive Granger himself put it: “the problem is how to devise a definition of causality and feedback and to test for their existence” (Granger 1969, p.428).

Regardless of the model dimensionality, it is possible to test the null of Granger non-causality if an asymptotically normal estimator is available. For illustration, consider an  $N$ -dimensional VAR(1) model. Given  $\mathcal{H} \subset [N] \times [N]$ , we may test  $H_0 : \phi_{ij} = 0$  for all  $(i, j) \in \mathcal{H}$  versus  $H_1 : \phi_{ij} \neq 0$  for some

$(i, j) \in \mathcal{H}$  using the asymptotic Wald statistic via the *debiased lasso* estimator (Zheng and Raskutti 2019; Zhu and Liu 2020; Babii, Ghysels, and Striaukas 2021). This is effective when a very small  $\mathcal{H}$  is of our interest. Unfortunately, however, this approach is not suitable for discovering the high-dimensional networks in  $\mathcal{H}$ . This is because, when  $\mathcal{H}$  is large, the rejection of this null hypothesis tells us only that there may exist Granger causality between *some* of the  $N$  variables. This is not informative for our purpose.

### 1.3. Discovering the Network with Directional FDR Control

The network discovering is characterized as the *multiple testing* for the sequence of hypotheses,  $H_0^{(ij)} : \phi_{ij} = 0$  versus  $H_1^{(ij)} : \phi_{ij} \neq 0$  for each  $(i, j) \in \mathcal{H}$ . Let  $\mathcal{S} = \{(i, j) : H_1^{(ij)} \text{ is true}\}$  and  $\hat{\mathcal{S}} = \{(i, j) : H_0^{(ij)} \text{ is rejected}\}$ . The *false discovery rate* (FDR), considered as a measure of Type I error in multiple tests, is defined as  $\text{FDR} = \mathbb{E}[|\hat{\mathcal{S}} \cap \mathcal{S}^c| / (|\hat{\mathcal{S}}| \vee 1)]$  (Benjamini and Hochberg 1995). The FDR-controlled multiple testing tends to exhibit higher power ( $\text{Power} = \mathbb{E}[|\hat{\mathcal{S}} \cap \mathcal{S}| / |\mathcal{S}|]$ ) than that controlling the *family-wise error rate* ( $\text{FWER} = \mathbb{P}(|\hat{\mathcal{S}} \cap \mathcal{S}^c| \geq 1)$ ), typically by the so-called Bonferroni correction (Bonferroni 1935; Holm 1979), because the FDR is smaller than or equal to the FWER. FDR control for time series is essentially difficult, and thus there are very few studies. Only exceptionally, Chi et al. (2021) develop a quite general knockoff framework for time series by extending Candès et al. (2018), but we do not pursue the direction.

In this article, we propose two novel methods for discovering the network Granger causality in high-dimensional VAR models, based on asymptotic and bootstrap  $t$ -statistics, respectively. We will show that they can control the *directional* FDR (dFDR), which takes into account not only whether each parameter is zero but also the differences in their directions (signs), with a high power guarantee. This will lead to a more accurate and stable discovery. In case a strong cross-sectional dependency in the model is anticipated, we also provide a *robustification* of Procedure 1 that gives a valid FDR control under any dependence structure. To the best of our knowledge, this is the first paper to propose such inference-based procedures that focus on discovering network Granger causalities with theoretical guarantees. The finite sample evidence suggests that the procedures can successfully control the FDR while maintaining the high power. The proposed methods are applied to a large dataset of

macroeconomic time series and regional house prices in the United Kingdom.

There are some studies that consider Granger causality between groups of variables, typically using group lasso; see Basu, Shojaie, and Michailidis (2015), Lin and Michailidis (2017), and Basu, Li, and Michailidis (2019). This approach requires the researcher to know the group members. Guðmundsson and Brownlees (2021) develop methods to statistically identify groups among the variables.

#### 1.4. Organization and Notation

The article is organized as follows. Section 2 defines the VAR model. Section 3 proposes two methods and a robustification technique for discovering the networks based on a multiple test. Section 4 explores the statistical theory for the FDR control and power guarantee of our methods. Section 5 confirms the finite sample validity via Monte Carlo experiments. Section 6 applies our methods to large datasets. Section 7 concludes. All the proofs and additional analyses are collected in supplementary material.

**Notation.** For any matrix  $\mathbf{M} = (m_{ti}) \in \mathbb{R}^{T \times N}$ , denote by  $\|\mathbf{M}\|_F$ ,  $\|\mathbf{M}\|_2$ ,  $\|\mathbf{M}\|_1$ ,  $\|\mathbf{M}\|_{\max}$ , and  $\|\mathbf{M}\|_\infty$  the Frobenius norm, induced  $\ell_2$  (spectral) norm, entrywise  $\ell_1$ -norm, entrywise  $\ell_\infty$ -norm, and induced  $\ell_\infty$ -norm, respectively. Specifically, they are defined by  $\|\mathbf{M}\|_F = (\sum_{t,i} m_{ti}^2)^{1/2}$ ,  $\|\mathbf{M}\|_2 = \lambda_1^{1/2}(\mathbf{M}'\mathbf{M})$ ,  $\|\mathbf{M}\|_1 = \sum_{t,i} |m_{ti}|$ ,  $\|\mathbf{M}\|_{\max} = \max_{t,i} |m_{ti}|$ , and  $\|\mathbf{M}\|_\infty = \max_t \sum_i |m_{ti}|$ , where  $\lambda_i(\mathbf{S})$  refers to the  $i$ th largest eigenvalue of a symmetric matrix  $\mathbf{S}$ . Denote by  $\mathbf{I}_N$  and  $\mathbf{0}_{T \times N}$  the  $N \times N$  identity matrix and  $T \times N$  matrix with all the entries being zero, respectively. We use  $\lesssim$  ( $\gtrsim$ ) to represent  $\leq$  ( $\geq$ ) up to a positive constant factor. For any positive sequence  $a_n$  and  $b_n$ , we write  $a_n \asymp b_n$  if  $a_n \lesssim b_n$  and  $a_n \gtrsim b_n$ . For any positive values  $a$  and  $b$ ,  $a \vee b$  and  $a \wedge b$  stand for  $\max(a, b)$  and  $\min(a, b)$ , respectively. The indicator function is denoted by  $1\{\cdot\}$ . The sign function is defined as  $\text{sgn}(x) = x/|x|$  for  $x \neq 0$  and  $\text{sgn}(0) = 0$ . The ceiling function is denoted as  $\lceil x \rceil = \min\{n \in \mathbb{Z} : n \geq x\}$ .

## 2. Model

Suppose that the  $N$ -dimensional vector of stationary time series,  $\mathbf{y}_t = (y_{1t}, \dots, y_{Nt})'$ , is generated from the VAR( $K$ ) model:

$$\mathbf{y}_t = \Phi_1 \mathbf{y}_{t-1} + \dots + \Phi_K \mathbf{y}_{t-K} + \mathbf{u}_t = \Phi \mathbf{x}_t + \mathbf{u}_t, \quad (1)$$

where  $\Phi_k \in \mathbb{R}^{N \times N}$  are the *sparse* coefficient matrices with  $\Phi_k = (\phi_{ij,k})$  and  $\Phi = (\Phi_1, \dots, \Phi_K)$ ,  $\mathbf{x}_t = (\mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-K})' \in \mathbb{R}^{KN}$  is the vector of lagged variables, and  $\mathbf{u}_t = (u_{1t}, \dots, u_{Nt})'$  is an error vector with mean zero and finite positive definite covariance matrix  $\Sigma_u = (\sigma_{ij}) = \mathbb{E} \mathbf{u}_t \mathbf{u}_t'$ . Let  $\Sigma_x = \mathbb{E} \mathbf{x}_t \mathbf{x}_t'$  and  $\Gamma_y(k-1) = \mathbb{E} \mathbf{y}_{t-1} \mathbf{y}'_{t-k}$  with  $\Gamma_y(k-1) = \Gamma_y(-k+1)'$  for  $k \in [K]$ . Then  $\Sigma_x$  is composed of submatrices  $\Gamma_y(k-1)$ . Moreover, define the precision matrix of  $\mathbf{x}_t$  as  $\Omega = (\omega_{ij}) = \Sigma_x^{-1}$ . Denote by  $\omega_j$  the  $j$ th column vector of  $\Omega$  for  $j \in [KN]$ . We follow the convention,  $\sigma_i^2 = \sigma_{ii}$  and  $\omega_i^2 = \omega_{ii}$ . Throughout the article, we suppose that  $N \wedge T \rightarrow \infty$  with allowing  $N$  to be as large as or possibly larger than  $T$  while  $K = o(N \wedge T)$ . We also assume  $K$  to be known for the sake of a concise presentation; see for example Nicholson et al. (2020) for a review and new lag selection method.

Stacking the  $T$  observations in columns such that  $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_T) \in \mathbb{R}^{N \times T}$ ,  $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T) \in \mathbb{R}^{KN \times T}$ , and  $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_T) \in \mathbb{R}^{N \times T}$ , we have the matrix form,  $\mathbf{Y} = \Phi \mathbf{X} + \mathbf{U}$ . Denote by  $\mathbf{a}_i$  the  $i$ th row vector of matrix  $\mathbf{A}$ . Then, by extracting the  $i$ th row of the matrix form, the model is also written as  $\mathbf{y}_i = \phi_i \mathbf{X} + \mathbf{u}_i$ .

To describe the sparsity pattern of  $\Phi$ , define  $S = \{(i, j) \in [N] \times [KN] : \phi_{ij} \neq 0\}$  with cardinality  $s = |S|$ . Similarly, the sparsity pattern of  $\phi_i$  is described as  $S_i = \{j \in [KN] : \phi_{ij} \neq 0\}$  with  $s_i = |S_i|$  and  $\bar{s} = \max_{i \in [N]} s_i$ . There is a one-to-one correspondence between  $S$  and the Granger-causal network as mentioned in Section 1.1. Our goal is to discover the network with controlling the FDR.

## 3. Inferential Methodology

We first propose two multiple testing procedures that can control the FDR of discovering the Granger-causal networks. They are based on asymptotic and bootstrap  $t$ -statistics, respectively. Then, we introduce robustified version of the first procedure that is valid under arbitrary cross-sectional dependence structure in the model. The associated statistical theory is developed in Section 4.

### 3.1. Debiased Lasso Estimator

We start with constructing the row-wise lasso estimator,  $\hat{\phi}_i^L$ , defined as

$$\hat{\phi}_i^L = \arg \min_{\phi_i \in \mathbb{R}^{1 \times KN}} (2T)^{-1} \|\mathbf{y}_i - \phi_i \mathbf{X}\|_2^2 + \lambda \|\phi_i\|_1, \quad (2)$$

where  $\lambda > 0$  is a regularization parameter. It is well-known that the lasso estimator has a bias caused by this regularization. Following Javanmard and Montanari (2014), we remove the bias, which leads to the asymptotic normality of each element; see also van de Geer et al. (2014) and Zhang and Zhang (2014) for a related discussion. Define the debiased lasso estimator as

$$\hat{\phi}_i = \hat{\phi}_i^L + (\mathbf{y}_i - \hat{\phi}_i^L \mathbf{X}) \mathbf{X}' \hat{\Omega} / T, \quad (3)$$

where  $\hat{\Omega}$  is a consistent precision matrix estimator. By a simple calculation, (3) is written as  $\sqrt{T}(\hat{\phi}_i - \phi_i) = \mathbf{z}_i + \mathbf{r}_i$  for some  $\mathbf{z}_i$  and  $\mathbf{r}_i$  such that for each  $i \in [N]$ , each element of  $\mathbf{z}_i$  can be asymptotically normal while  $\|\mathbf{r}_i\|_{\max}$  becomes negligible under regularity conditions. This point is formally investigated in Section 4.1.

### 3.2. Multiple Test

For any  $\mathcal{H} \subset [N] \times [KN]$ , consider discovering the Granger-causal network in  $\mathcal{H}$ . This problem is understood as the multiple test for the sequence of hypotheses:

$$H_0^{(i,j)} : \phi_{ij} = 0 \text{ versus } H_1^{(i,j)} : \phi_{ij} \neq 0 \text{ for } (i, j) \in \mathcal{H}. \quad (4)$$

As observed in Section 3.1, under regularity conditions, the debiased lasso estimator will be expressed as  $\sqrt{T} \hat{\phi}_{ij} = z_{ij} + o_p(1)$  under  $H_0^{(i,j)}$ , where  $z_{ij}$  is expected to be asymptotically normal

with the variance,  $\text{var}(z_{ij}) = \sigma_i^2 \omega_j' \Sigma_x \omega_j = \sigma_i^2 \omega_j^2$ . Thus, the  $t$ -test for each pair of hypotheses in (4) is performed with the  $t$ -statistic, either

$$T_{ij} = \frac{\sqrt{T} \hat{\phi}_{ij}}{\hat{\sigma}_i \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}} \quad \text{or} \quad \frac{\sqrt{T} \hat{\phi}_{ij}}{\hat{\sigma}_i \hat{\omega}_j}, \quad (5)$$

where  $\hat{\Sigma}_x = \mathbf{X}\mathbf{X}'/T$ . Hereafter, denote by  $\hat{m}_{ij}$  either  $\hat{\sigma}_i \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}$  or  $\hat{\sigma}_i \hat{\omega}_j$ .

Repeating the  $t$ -test over  $(i, j) \in \mathcal{H}$  with a critical value  $\tau$  leads to a set of discoveries,  $\hat{\mathcal{S}}(\tau) := \{(i, j) \in \mathcal{H} : |T_{ij}| \geq \tau\}$ . Here, the choice of  $\tau$  is critical; we propose two procedures to determine  $\tau$  that achieves controlling the *directional FDR* of  $\hat{\mathcal{S}}(\tau)$ ,

$$\text{dFDR} = \mathbb{E}[\text{dFDP}],$$

$$\text{dFDP} = \frac{| \{(i, j) \in \hat{\mathcal{S}}(\tau) : \text{sgn}(\hat{\phi}_{ij}) \neq \text{sgn}(\phi_{ij}) \} |}{|\hat{\mathcal{S}}(\tau)| \vee 1},$$

below a target level. The dFWER and dPower are also defined in a similar manner.

There are several ways to construct consistent estimators of the nuisance parameters,  $\sigma_i^2$  and  $\omega_j$ . In this article, we use  $\hat{\sigma}_i^2 = \sum_{t=1}^T \hat{u}_{it}^2 / (T - d_i)$  with  $\hat{u}_{it}$  the  $(i, t)$ th element of the lasso residual matrix  $\hat{\mathbf{U}}$  and the CLIME estimator (Cai, Liu, and Luo 2011)  $\hat{\boldsymbol{\Omega}} = (\hat{\omega}_1, \dots, \hat{\omega}_{KN})$ , respectively. Here,  $d_i = o(T)$  is a positive number for degrees of freedom adjustment. A typical choice is  $d_i = \hat{s}_i$ , where  $\hat{s}_i = |\hat{\mathcal{S}}_i^L|$  with  $\hat{\mathcal{S}}_i^L = \text{supp}(\hat{\boldsymbol{\phi}}_i^L)$ .

### 3.2.1. First Procedure: Limiting Normal Distribution

We construct the set of discoveries as follows.

**Procedure 1.** 1. Set  $\bar{\epsilon} = \sqrt{2 \log(|\mathcal{H}|) - a \log \log(|\mathcal{H}|)}$  for given  $\mathcal{H} \subset [N] \times [KN]$ , where  $a > 3$  is an arbitrary fixed constant.  
2. For any level  $q \in [0, 1]$ , compute

$$\tau_0 = \inf \left\{ \tau \in [0, \bar{\epsilon}] : \frac{2|\mathcal{H}|Q(\tau)}{|\hat{\mathcal{S}}(\tau)| \vee 1} \leq q \right\}, \quad (6)$$

where  $|\hat{\mathcal{S}}(\tau)| = \sum_{(i,j) \in \mathcal{H}} 1\{|T_{ij}| \geq \tau\}$  is the total number of discoveries in  $\mathcal{H}$  with critical value  $\tau$ , and  $Q(\tau) = \mathbb{P}(\mathcal{Z} > \tau)$  with  $\mathcal{Z}$  a standard normal random variable. If (6) does not exist, set

$$\tau_0 = \sqrt{2 \log(|\mathcal{H}|)}. \quad (7)$$

3. Obtain the set of discoveries,  $\hat{\mathcal{S}}(\tau_0) = \{(i, j) \in \mathcal{H} : |T_{ij}| \geq \tau_0\}$ .

This procedure is designed to asymptotically control the dFDR of rejected nulls,  $\hat{\mathcal{S}}(\tau_0)$ , to be less than or equal to  $q$ . A similar procedure can be found in Liu (2013), Javanmard and Javadi (2019), and Uematsu and Yamagata (2023), for example. The dFDR control always implies the FDR control since  $\text{FDR} \leq \text{dFDR}$ . Threshold (7) will even control the dFWER (and hence dFDR), which occurs if it is large enough not to exist in  $[0, \bar{\epsilon}]$ . A theoretical justification for the dFDR control and dPower guarantee is given in Section 4.2.

**Remark 1.** (a) The constant  $a > 3$  is required for technical reasons. In practice, we can choose an arbitrary value that is slightly larger than three, like  $a = 3.001$ . This is not sensitive to the selection result as long as it is sufficiently small.

(b) A choice of  $d_i$  effects the FDR control to some extent, but  $d_i = \hat{s}_i$  seems to bring good results by simulation studies in Section 5. For other choices of  $d_i$  and constructions of consistent estimator of  $\sigma_i^2$ , see Reid, Tibshirani, and Friedman (2016).

### 3.2.2. Second Procedure: Bootstrapped Distribution

Even in a low-dimensional setting, Brüggemann, Jentsch, and Trenkler (2016) point out that the finite sample properties of asymptotic VAR inference can be rather poor, and the use of bootstrap methods is often advocated. See also Kilian (1999), Gonçalves and Kilian (2004), Hafner and Herwartz (2009). To improve the performance, we propose a second method based on the *fixed-design wild bootstrap* (FWB). This is different from a conventional bootstrapped  $t$ -test for a single hypothesis. Roughly speaking, it attempts to replicate a bootstrapped distribution under the null by the FWB, which is then substituted for the limit normal distribution  $Q$  in Procedure 1.

Let  $\tilde{\mathcal{S}} \subset [N] \times [KN]$  denote an index set of discoveries that is expected to satisfy  $\mathbb{P}(\tilde{\mathcal{S}} \supset \mathcal{S}) \rightarrow 1$ . A typical choice of  $\tilde{\mathcal{S}}$  is the lasso-selected variables,  $\hat{\mathcal{S}}_L$ . We may also use  $\hat{\mathcal{S}}(\tau_0)$  obtained by Procedure 1.

**Procedure 2.** 1. Obtain a bootstrap version of the  $t$ -statistics,  $\{T_{ij}^{*(b)} : (i, j) \in \tilde{\mathcal{S}}^c \cap \mathcal{H}\}_{b=1}^B$ , by repeating (a)–(f)  $B$  times:

- Generate  $\{\zeta_t\}$ , a sequence of iid random variables with mean zero and variance one.
- Obtain  $\{\mathbf{u}_t^*\}_{t=1}^T$  by the FWB; that is,  $\mathbf{u}_t^* = \hat{\mathbf{u}}_t \zeta_t$ , where  $\hat{\mathbf{u}}_t = \mathbf{y}_t - \hat{\boldsymbol{\Phi}}^L \mathbf{x}_t$ .
- Generate  $\{\mathbf{y}_t^*\}_{t=1}^T$  by  $\mathbf{y}_t^* = \hat{\boldsymbol{\Phi}}^L \mathbf{x}_t + \mathbf{u}_t^*$ , and set  $\mathbf{Y}^* = (\mathbf{y}_1^*, \dots, \mathbf{y}_T^*)$ .
- Compute a bootstrap version of the lasso estimate  $\hat{\boldsymbol{\Phi}}^{L*}$  by  $\mathbf{Y}^*$  and  $\mathbf{X}$ .
- Construct a bootstrap version of the debiased lasso estimate,  $\hat{\boldsymbol{\Phi}}^* = \hat{\boldsymbol{\Phi}}^{L*} + (\mathbf{Y}^* - \hat{\boldsymbol{\Phi}}^{L*} \mathbf{X}) \mathbf{X}' \hat{\boldsymbol{\Omega}} / T$ .
- Construct a bootstrap version of the  $t$ -statistics, either

$$T_{ij}^* = \frac{\sqrt{T} \hat{\phi}_{ij}^*}{\hat{\sigma}_i^* \sqrt{\hat{\omega}_j^{*'} \hat{\Sigma}_x \hat{\omega}_j^*}} \quad \text{or} \quad \frac{\sqrt{T} \hat{\phi}_{ij}^*}{\hat{\sigma}_i^* \hat{\omega}_j^*} \quad \text{for } (i, j) \in \tilde{\mathcal{S}}^c \cap \mathcal{H}, \quad (8)$$

where  $\hat{\sigma}_i^{*2}$  is given by  $\hat{\sigma}_i^{*2} = \sum_{t=1}^T \hat{u}_{it}^{*2} / (T - \hat{s}_i)$  with  $\hat{\mathbf{U}}^* = \mathbf{U}^* - (\hat{\boldsymbol{\Phi}}^{L*} - \hat{\boldsymbol{\Phi}}^L) \mathbf{X}$  and  $\mathbf{U}^* = (\mathbf{u}_1^*, \dots, \mathbf{u}_T^*)$ .

2. Compute the empirical distribution,

$$\mathbb{Q}_B^*(\tau) = \frac{1}{|\tilde{\mathcal{S}}^c \cap \mathcal{H}|} \sum_{(i,j) \in \tilde{\mathcal{S}}^c \cap \mathcal{H}} \left[ \frac{1}{B} \sum_{b=1}^B 1\{T_{ij}^{*(b)} > \tau\} \right].$$

3. Run Procedure 1 with (9) replacing (6):

$$\tau_0 = \inf \left\{ \tau \in [0, \bar{\epsilon}] : \frac{|\mathcal{H}| \{ \mathbb{Q}_B^*(\tau) + 1 - \mathbb{Q}_B^*(-\tau) \}}{|\hat{\mathcal{S}}(\tau)| \vee 1} \leq q \right\}. \quad (9)$$



**Remark 2.** (a) We may adopt several distributions for  $\zeta_t$ . For example, Mammen (1993) suggests using  $\zeta_t$  that takes values  $\pm(\sqrt{5}\pm 1)/2$  with probability  $(\sqrt{5}\pm 1)/(2\sqrt{5})$ , respectively, while Davidson and Flachaire (2008) propose using the Rademacher random variable,  $\zeta_t = \pm 1$  with probability  $1/2$ .

(b) Unlike the construction of  $\hat{\sigma}_i^2$  in Procedure 1, the degree of freedom adjustment in  $\hat{\sigma}_i^{*2}$  is not sensitive to the results of FDR control.

### 3.2.3. Robustifying Transformation

The (directional) FDR control by the proposed two procedures requires a condition to restrict the correlation structure of  $z_{ij}$ 's; see Condition 6 in the next section. Such a condition that is somewhat restrictive and difficult to verify if actually met in real data often tends to be avoided. As a remedy, we propose *robustification* of Procedure 1, which allows any cross-sectional dependency in  $z_{ij}$ 's. As a tradeoff for the robustness, it exhibits conservative selection in general. Thus, the procedure is recommended when a cautious analysis is needed.

The key is to transform the  $t$ -statistics to an  $e$ -variables, the definition of which is a nonnegative random variable  $E$  with  $\mathbb{E}[E] \leq 1$  under the null (Vovk and Wang 2021). The significant factor is that  $1/E$  becomes a  $p$ -variable since  $\mathbb{P}(1/E \leq q) \leq q$  for any level  $q \geq 0$  by the Markov inequality, and is valid under any dependence structure. Then a robust FDR-controlled selection is achieved via the standard BH procedure of Benjamini and Hochberg (1995) with  $1/E$ , which is called the  $e$ -BH (Wang and Ramdas 2022).

To introduce the  $e$ -BH procedure, we convert the double index  $(i, j) \in \mathcal{H}$  to an ordered single index  $h \in [|\mathcal{H}|]$  using some converter  $h(i, j)$ . For example, if  $\mathcal{H} = [N] \times [KN]$ , we set  $h = h(i, j)$  with  $h(i, j) = i + N(j - 1)$  for given  $(i, j) \in \mathcal{H}$ ; conversely,  $(i, j)$  is recovered by setting  $j = \lceil h/N \rceil$  and  $i = h - N(j - 1)$  for given  $h \in [|\mathcal{H}|]$ . Suppose that  $|\mathcal{H}|$   $e$ -variables  $E_h = E_{h(i, j)}$  corresponding to  $H_0^{(i, j)}$  are available.

**Procedure 3 ( $e$ -BH).** 1. Make order statistics,  $E_{(1)} \geq \dots \geq E_{(|\mathcal{H}|)}$ .  
2. Compute  $h^* = \max \{h \in [|\mathcal{H}|] : 1/E_{(h)} \leq qh/|\mathcal{H}|\}$ .  
3. Obtain the set of discoveries,  $\hat{S}_R(h^*) = \{(i, j) \in \mathcal{H} : h(i, j) = h, j = \lceil h/N \rceil, E_h \geq E_{(h^*)}\}$ .

A critical aspect is how to construct a “good”  $e$ -variable as there are countless ways to make it. Since little is known about the optimal construction, we focus on a simple formulation of  $e$ -variables transformed from the  $t$ -statistics (5), which we call the *robustifying transformation*. Precisely, we adopt the following:

$$E_{h(i, j)} = \frac{f(T_{ij})}{\mathbb{E}[f(\mathcal{Z})]}, \quad \mathcal{Z} \sim N(0, 1), \quad (10)$$

where  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  is a continuous strictly increasing function. We may compute  $\mathbb{E}[f(\mathcal{Z})]$  once  $f$  is specified. This construction yields asymptotic  $e$ -variables since  $\mathbb{E}[E_{h(i, j)}] \rightarrow 1$  is expected under  $H_0^{(i, j)}$ ; whereas, under  $H_1^{(i, j)}$ , we can obtain  $\mathbb{E}[E_{h(i, j)}] \asymp f(\sqrt{T}) \rightarrow \infty$  thanks to the fact of  $T_{ij} \approx \sqrt{T}$  and increasing  $f$ . In practice, we may use  $f : x \mapsto |x|^p$  for some  $p > 0$  with  $\mathbb{E}|\mathcal{Z}|^p = \sqrt{2^p/\pi} \Gamma((p+1)/2)$  or  $f : x \mapsto \exp(c|x|^\alpha)$

for some  $c, \alpha > 0$  with  $\mathbb{E}[\exp(c|\mathcal{Z}|^\alpha)] = 2 \exp(c^2/2) \Phi(c)$  if  $\alpha = 1$ . A theory is provided in Section 4.2.3 and Section E of supplementary material.

## 4. Statistical Theory

We develop a formal statistical theory for the inferential methodology proposed in Section 3. Throughout this section, we set  $\mathcal{H} = [N] \times [KN]$ ,  $d_i = 0$ , and  $\tilde{S} = \hat{S}_L$  to alleviate unnecessary technical complications. We suppose that  $N, T \rightarrow \infty$ ,  $N = O(T^d)$  for some  $d > 0$ , and  $K = o(N \vee T)$ .

**Condition 1.** The error term,  $\{\mathbf{u}_t\}$ , is a sequence of iid sub-Gaussian random vectors with mean zero and covariance matrix  $\Sigma_u$ ; for every  $N > 0$ , there exists some constant  $c_u > 0$  such that for all  $x > 0$ ,  $\max_{i \in [N]} \mathbb{P}(|u_{it}| > x) \leq 2 \exp(-x^2/c_u)$ .

**Condition 2.** All the eigenvalues of the companion matrix of  $(\Phi_1, \dots, \Phi_K)$  are strictly less than one in modulus uniformly in  $K$  and  $N$ .

**Condition 3.** For all  $K$  and  $N$ , there exists some constant  $\gamma > 0$  such that  $\gamma \leq \lambda_{\min}(\Sigma_x) \leq \lambda_{\max}(\Sigma_x) \leq 1/\gamma$ .

Conditions 1–3 are commonly used in the literature. Condition 2 guarantees that the VAR( $K$ ) model is stable and is inverted to the VMA( $\infty$ ) model:

$$\mathbf{y}_t = \sum_{\ell=0}^{\infty} \mathbf{B}_\ell \mathbf{u}_{t-\ell}, \quad b := \sum_{\ell=0}^{\infty} \|\mathbf{B}_\ell\|_\infty < \infty, \quad (11)$$

where  $\mathbf{B}_0 = \mathbf{I}_N$  and  $\mathbf{B}_\ell = \mathbf{J}' \mathbf{A}^\ell \mathbf{J}$  with  $\mathbf{J}' = (\mathbf{I}_N, \mathbf{0}_{N \times (KN-N)})$  for  $\ell = 1, 2, \dots$  and  $\mathbf{A}$  the companion matrix of  $(\Phi_1, \dots, \Phi_K)$ ; see Lemma 2 in supplementary material. Throughout this section, we set the lasso regularization parameter in (2) to be

$$\lambda = 8bc_{uu} \sqrt{2(\nu+7)^3 T^{-1} \log^3(N \vee T)},$$

where  $\nu > 0$  is a fixed constant.

### 4.1. Theory for the Debiased Lasso Estimator

The first proposition derives the nonasymptotic error bounds of the lasso estimator.

**Proposition 1 (Nonasymptotic error bounds for the lasso).** If Conditions 1–3 hold, then the lasso estimator defined in (2) satisfies the following inequalities with probability at least  $1 - O((N \vee T)^{-\nu})$ :

$$\begin{aligned} (a) \quad & \left\| \hat{\phi}_i^L - \phi_i \right\|_2 \leq \frac{12\sqrt{s_i}\lambda}{\gamma - 8bs_i\lambda}, \\ (b) \quad & \left\| \hat{\phi}_i^L - \phi_i \right\|_1 \leq \frac{48s_i\lambda}{\gamma - 8bs_i\lambda}, \\ (c) \quad & \frac{1}{T} \left\| (\hat{\phi}_i^L - \phi_i) \mathbf{X} \right\|_2^2 \leq \frac{144s_i\lambda^2}{\gamma - 8bs_i\lambda} \end{aligned}$$

for all  $i \in [N]$  such that  $8bs_i\lambda < \gamma$ .

Using [Proposition 1](#), we next show the asymptotic linearity of the debiased lasso estimator,  $\hat{\Phi}$ . This requires some conditions on the precision matrix,  $\Omega$ .

**Condition 4.** There exist some positive numbers  $s_\omega$  and  $M_\omega$  and some constant  $r \in [0, 1)$  such that  $\max_{i \in [KN]} \sum_{j=1}^{KN} |\omega_{ij}|^r \leq s_\omega$  and  $\max_{j \in [KN]} \|\omega_j\|_1 \leq M_\omega$ , where  $s_\omega$  and  $M_\omega$  can diverge as  $N, T \rightarrow \infty$ .

[Condition 4](#) has frequently been used to derive the rate of convergence of the CLIME estimator  $\hat{\Omega}$ ; see for example, Cai, Liu, and Luo (2011) and Shu and Nan (2019). This condition requires that  $\Omega$  is approximately sparse. Especially when  $r = 0$ , the condition reduces to the exact sparsity assumption. Denote by  $\{e_j\}$  the standard basis of  $\mathbb{R}^{KN}$ .

**Proposition 2 (Asymptotic linearity of the debiased lasso).** The debiased lasso estimator defined in (3) has the representation  $\sqrt{T}(\hat{\Phi} - \Phi) = Z + R$ , where the  $(i, j)$ th elements of  $Z$  and  $R$  are respectively given by

$$z_{ij} = \frac{1}{\sqrt{T}} \sum_{t=1}^T u_{it} \mathbf{x}'_t \omega_j,$$

$$r_{ij} = \frac{1}{\sqrt{T}} \mathbf{u}_i \mathbf{X}' (\hat{\omega}_j - \omega_j) - \sqrt{T} (\hat{\phi}_i^L - \phi_i) (\hat{\Sigma}_x \hat{\omega}_j - e_j).$$

Furthermore, if [Conditions 1–4](#) are assumed, then the following inequality holds with probability at least  $1 - O((N \vee T)^{-\nu})$ :

$$\max_{j \in [KN]} |r_{ij}| \lesssim (s_\omega M_\omega^{2-2r} \lambda^{1-r} + M_\omega s_i \lambda) \sqrt{\log^3(N \vee T)} =: \bar{r}_i \quad (12)$$

for all  $i \in [N]$  such that  $s_i \lambda = o(1)$ .

To achieve the asymptotic normality of  $\sqrt{T}(\hat{\phi}_{ij} - \phi_{ij})$ , we need  $\bar{r}_i = o(1)$ . Then  $z_{ij}$  becomes the dominating term, which can converge in distribution to a Gaussian random variable. Note that  $\bar{r}_i \asymp (s_\omega^3 + s_\omega s_i) \ell_{NT} / \sqrt{T}$ , where  $\ell_{NT}$  is some power of  $\log(N \vee T)$ , if  $r = 0$  in [Condition 4](#).

## 4.2. Theory for the Multiple Testing

We are now ready to develop a statistical theory of the multiple testing of (4) by [Procedures 1–3](#) in [Section 3.2](#). We first establish the asymptotic normality of the  $t$ -statistics defined in (5).

**Condition 5.** For all  $K$  and  $N$ , there exists some constant  $\gamma > 0$  such that  $\gamma \leq \min_{i \in [N]} \sigma_i^2 \leq \max_{i \in [N]} \sigma_i^2 \leq 1/\gamma$  and  $\gamma \leq \min_{j \in [KN]} \omega_j^2 \leq \max_{j \in [KN]} \omega_j^2 \leq 1/\gamma$ .

[Condition 5](#) complements [Condition 3](#), and is required to deal with the standard errors; see Lemma 7 in supplementary material.

**Theorem 1 (Asymptotic normality of  $t$ -statistic).** If [Conditions 1–5](#) hold, then either  $t$ -statistics  $T_{ij}$  defined in (5) satisfies that  $T_{ij} - \sqrt{T} \phi_{ij} / \hat{m}_{ij}$  converges in distribution to  $N(0, 1)$  for all  $(i, j) \in \mathcal{H}$  such that  $\bar{v}_i := \bar{r}_i + M_\omega^2 \lambda = o(1)$ .

In the theorem,  $\bar{v}_i = o(1)$  makes both  $\bar{r}_i$  in (12) and the estimation error of the standard error,  $M_\omega^2 \lambda$ , asymptotically negligible. If  $r = 0$  in [Condition 4](#), it reduces to  $(s_\omega^6 + s_\omega^2 s_i^2) \ell_{NT} \ll T$ .

### 4.2.1. Theory for the FDR Control

For  $((i, j), (k, \ell)) \in \mathcal{H} \times \mathcal{H} =: \mathcal{H}^2$ , the correlation between  $z_{ij}$  and  $z_{k\ell}$  is given by

$$\rho_{(i,j),(k,\ell)} := \text{Corr}(z_{ij}, z_{k\ell}) = \frac{\sigma_{ik} \omega_{j\ell}}{\sigma_i \sigma_k \omega_j \omega_\ell}, \quad (13)$$

which becomes  $|\rho_{(i,j),(k,\ell)}| \asymp |\sigma_{ik} \omega_{j\ell}|$  under [Condition 5](#). Obviously, we have  $\rho_{(i,j),(i,j)} = 1$  for all  $(i, j) \in \mathcal{H}$ , but the “off-diagonal” correlations in

$$\mathcal{H}_{\text{off}}^2 := \{((i, j), (k, \ell)) \in \mathcal{H}^2 : (i, j) \neq (k, \ell)\}$$

with  $|\mathcal{H}_{\text{off}}^2| = |\mathcal{H}^2| - |\mathcal{H}|$  take nontrivial values. To manipulate them, impose the following condition.

**Condition 6.** There exists a partition of  $\mathcal{H}_{\text{off}}^2$  denoted as  $\mathcal{H}_{\text{off}}^2 = \mathcal{H}_w^2 \cup \mathcal{H}_s^2$  such that for some constant  $c > 0$  and  $\bar{\rho} := 1 - \varepsilon$  with arbitrary constant  $\varepsilon \in (0, 1)$ ,

$$|\rho_{(i,j),(k,\ell)}| \in \begin{cases} [0, c/(\log N)^2] & \text{for } ((i, j), (k, \ell)) \in \mathcal{H}_w^2 \\ & \text{(weak correlations),} \\ (c/(\log N)^2, \bar{\rho}] & \text{for } ((i, j), (k, \ell)) \in \mathcal{H}_s^2 \\ & \text{(strong correlations),} \end{cases}$$

where  $|\mathcal{H}_w^2| = |\mathcal{H}_{\text{off}}^2| - |\mathcal{H}_s^2|$  and  $|\mathcal{H}_s^2| = O(|\mathcal{H}_{\text{off}}^2|/(\log N)^2)$ .

[Condition 6](#) says that most of the correlations are “weak” so that they are close to zero, while some are allowed to be “strong” enough to be non-vanishing. Because  $T_{ij}$ ’s are asymptotically normal as shown in [Theorem 1](#), [Condition 6](#) ensures that most of  $T_{ij}$ ’s are asymptotically independent. This condition can still be satisfied by many VAR models. To see this, note that  $|\rho_{(i,j),(k,\ell)}| \asymp |\sigma_{ik} \omega_{j\ell}|$  and  $\sigma_{ik}, \omega_{j\ell} = O(1)$  under [Condition 5](#). Therefore, for example, if either  $\Sigma_u = (\sigma_{ik})$  or  $\Omega = (\omega_{j\ell})$  is approximately sparse, [Condition 6](#) can be satisfied regardless of the structure of the other. Here, neither  $\Sigma_u$  nor  $\Omega$  needs to be exactly sparse.

**Theorem 2 (FDR control: normal distribution).** Suppose  $\nu > 4$  and  $\max_{i \in [N]} \bar{v}_i = O(T^{-\kappa_1})$  for some constant  $\kappa_1 \in (0, 1/2)$ . If [Conditions 1–6](#) hold, then for any predetermined level  $q \in [0, 1]$ , [Procedure 1](#) with either  $t$ -statistics in (5) achieves the following: If  $\mathfrak{t}_0$  is given by (6), the obtained  $\hat{S}(\mathfrak{t}_0)$  satisfies  $\limsup_{N, T \rightarrow \infty} \text{dFDR} \leq q$  and  $\lim_{N, T \rightarrow \infty} \mathbb{P}(\text{dFDP} \leq q + \varepsilon) = 1$  for any  $\varepsilon > 0$ . If  $\mathfrak{t}_0$  is given by (7), the obtained  $\hat{S}(\mathfrak{t}_0)$  satisfies  $\limsup_{N, T \rightarrow \infty} \text{dFWER} \leq q$ .

Recall  $\bar{v}_i = \bar{r}_i + M_\omega^2 \lambda$  defined in [Theorem 2](#). The condition,  $\max_{i \in [N]} \bar{v}_i = O(T^{-\kappa_1})$ , reduces to  $(s_\omega^6 + s_\omega^2 \bar{s}^2)^{1/(1-2\kappa_1)} \ell_{NT} \ll T$ , where  $\bar{s} = \max_{i \in [N]} s_i$ , if  $r = 0$  in [Condition 4](#).

We next show the dFDR control of [Procedure 2](#) with setting  $\tilde{S} = \hat{S}_L$ . For this purpose, we need a distributional assumption on the wild bootstrap.

**Condition 7.**  $\{\zeta_t\}$  is a sequence of iid sub-Gaussian random variables with  $\mathbb{E} \zeta_t = 0$  and  $\mathbb{E} \zeta_t^2 = 1$ , where for every  $T > 0$ , there exists some constant  $c_\zeta > 0$  such that for all  $x > 0$ ,  $\max_{t \in [T]} \mathbb{P}(|\zeta_t| > x) \leq 2 \exp(-x^2/c_\zeta)$ .

Define the bootstrap distribution as

$$\mathbb{Q}^*(t) = \frac{1}{|\hat{\mathcal{S}}_L^c|} \sum_{(i,j) \in \hat{\mathcal{S}}_L^c} \mathbb{P}^* \left( T_{ij}^* > t \right) \text{ with}$$

$$\mathbb{P}^*(T_{ij}^* > t) = \text{plim}_{B \rightarrow \infty} \frac{1}{B} \sum_{b=1}^B 1\{T_{ij}^{*(b)} > t\},$$

where  $|\hat{\mathcal{S}}_L^c| = KN^2 - \hat{s}$ . Let  $\bar{\mu} = \max_{i \in [N]} \bar{v}_i + s_\omega M_\omega^{3-2r} \lambda^{1-r} \log^2(N \vee T)$ .

**Theorem 3 (FDR control: Bootstrap).** Suppose  $\nu > 4$  and  $\bar{\mu} = O(T^{-\kappa_1})$  for some constant  $\kappa_1 \in (0, 1/2)$ . If Conditions 1–7 hold, then for any preassigned level  $q \in [0, 1]$ , Procedure 2 with either  $t$ -statistics in (8) achieves the same results as Theorem 2.

The condition,  $\bar{\mu} = O(T^{-\kappa_1})$ , reduces to  $(s_\omega^8 + s_\omega^2 \bar{s}^2)^{1/(1-2\kappa_1)} \ell_{NT} \ll T$  if  $r = 0$  in Condition 4.

To achieve Theorem 3, we should show that  $t_0$  using  $\mathbb{Q}^*$  is asymptotically the same as using  $Q$ . Toward this goal, it suffices to verify that the event,  $|\mathbb{Q}^*(t)/Q(t) - 1|$  and  $|\{1 - \mathbb{Q}^*(-t)\}/Q(t) - 1|$  converge to zero uniformly in  $t \in [0, \bar{t}]$ , occurs with high probability. This proof will complete in two steps. First, show that  $T_{ij}^*$  with the “sandwich” s.e. can be approximated by the self-normalized sum,  $\sum_{i=1}^T \hat{u}_{it}^* \mathbf{x}_i' \hat{\omega}_j / \sum_{i=1}^T (\hat{u}_{it}^* \mathbf{x}_i' \hat{\omega}_j)^2$ . Then, verify that it can be uniformly normally approximated in the relative error using the Cramér-type large deviation theory of Jing, Shao, and Wang (2003). The additional cost  $s_\omega M_\omega^{3-2r} \lambda^{1-r} \log^2(N \vee T)$  in  $\bar{\mu}$  is due to the approximation error in the first step.

#### 4.2.2. Theory for the Power Guarantee

We next investigate the asymptotic power of our Procedures 1 and 2. A condition on the signal strength is required to distinguish the nonzero elements from zeros.

**Condition 8.** For  $\mathcal{S} \subset \mathcal{H}$ , it holds that  $\min_{(i,j) \in \mathcal{S}} |\phi_{ij}| / (\sigma_i \omega_j) \geq 4\sqrt{2 \log(KN^2)/T}$ .

**Theorem 4 (Power guarantee).** Suppose  $\nu > 4$  and  $\max_{i \in [N]} \bar{v}_i = O(T^{-\kappa_1})$  for some constant  $\kappa_1 \in (0, 1/2)$ . If Conditions 1–5 and 8 hold, then both Procedures 1 and 2 achieve  $\lim_{N,T \rightarrow \infty} \text{dPower} = 1$ .

This guarantees that both Procedures 1 and 2 do not asymptotically miss any important relation between variables, whichever threshold of either (6) or (7) is selected.

#### 4.2.3. Theory for the Robustification

As noted in Section 3.2.3, Condition 6 sometimes seems annoying though it is crucial for Procedures 1 and 2. Even without the condition, Procedure 3 with transformed statistics (10) can still provide reasonable discoveries.

**Theorem 5.** If Conditions 1–5 and  $\bar{v} = o(1)$  hold. For a given level  $q \in [0, 1]$ , the set of discoveries,  $\hat{\mathcal{S}}_R(h^*)$ , obtained by Procedure 3 with  $e$ -variables (10) satisfies the following:

- For any continuous function  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  such that  $f(T_{ij})$  is uniformly integrable (UI) for every  $(i, j) \in \mathcal{S}^c$ , we have  $\limsup_{N,T \rightarrow \infty} \text{FDR} \leq q$ .
- For any monotonically increasing continuous function  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  such that  $f(\sqrt{T})/|\mathcal{H}| \rightarrow \infty$  for every  $(i, j) \in \mathcal{S}$ , we have  $\lim_{N,T \rightarrow \infty} \text{Power} = 1$ .

It is not difficult to choose  $f$  that admits the condition of (b). As for (a), the UI of  $f(T_{ij})$  (as a random sequence indexed by  $(T, N, K)$  for every  $(i, j) \in \mathcal{S}^c$ ) is pivotal for our  $e$ -based robust FDR control. For instance, we can show that  $|T_{ij}|^p$  and  $\exp(c|T_{ij}|^\alpha)$  are UI for some constants  $p, c, \alpha > 0$ ; see Section E of supplementary material.

The theory for  $e$ -based robustification still has room for exploration. First, Theorem 5 does not investigate the directional FDR and power. Second, an optimal functional form of  $f$  is not well understood. These issues will be addressed as future research topics.

## 5. Monte Carlo Experiments

We investigate the finite sample behavior of the proposed procedures.

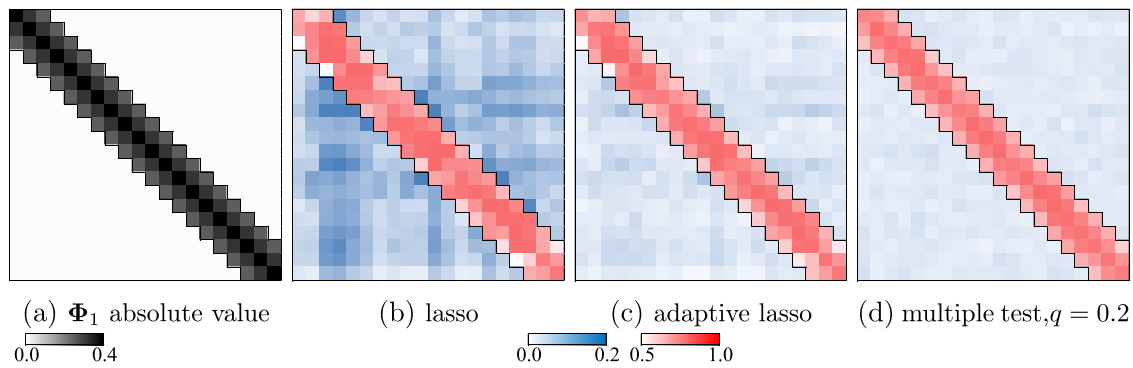
### 5.1. First Experiment: Stability

We begin with a small experiment to illustrate the superiority of our approach to those by the (adaptive) lasso in terms of selection stability. An  $N \times 1$  vector  $\mathbf{y}_t$  for  $t = -50, \dots, T$  is generated from the stationary VAR(1) model for  $N = 20$  and  $T = 100$  with  $\mathbf{u}_t \sim \text{iid}N(\mathbf{0}, \mathbf{I}_N)$ . The sparse  $\Phi$  is generated as in Section 5.2.1 in such a way that the diagonal and  $j$ -diagonals for  $j = \pm 1, \pm 2$ , are nonzero.

Figure 3 shows heatmaps of the frequencies with which the  $(i, j)$ th element of  $\Phi$  is discovered as nonzero by different methods in 1000 replications. Figure 3(a) shows a heatmap of the absolute values of elements in  $\Phi_1$ , and Figures 3(b)–(d) report the results using the lasso, adaptive lasso, and Procedure 1 with the target FDR level  $q = 0.2$ , respectively. The overall frequency of false positives by the lasso is very high, and for certain elements unreasonably so (shown by dark blue). The adaptive lasso reduces overall false positives, but cannot do so sufficiently for the elements wrongly selected in large numbers by lasso. In addition, some nonzero  $\phi_{ij}$ 's are selected too infrequently by lasso, which is largely inherited by adaptive lasso (shown by very weak red). Meanwhile, when Procedure 1 is used, the frequency of false positives is evenly spread among the elements and well controlled. Furthermore, nonzero elements are selected more evenly than the (adaptive) lasso. These suggest that the proposed inferential methods can provide more stable and reproducible results for discovering the network Granger causality than existing popular estimation methods.

Another practical advantage of our inference method is that by varying the value of  $q$ , elements of  $\Phi$  with different levels of significance can be visualized. By decreasing the value of  $q$ , the overall rejection frequency naturally decreases. In particular, the smaller the value of  $q$ , the more statistically significant elements are more likely to be selected; see the results with  $q = 0.1$





**Figure 3.** Heatmap of selection frequencies in  $\Phi_1$ ; the blue and red cells indicate the frequencies of false and true discoveries, respectively.

in Figure F1 in supplementary material. Clearly, point estimation methods, such as the (adaptive) lasso, cannot provide this feature.

## 5.2. Second Experiment: Performance of Our Methods

### 5.2.1. Design

We consider the  $N \times 1$  vector generated as  $\mathbf{y}_t = \sum_{\ell=1}^K \Phi_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t$  for  $t = -(50 + K), \dots, T$  with  $\mathbf{y}_{-(50+K)} = \mathbf{0}$ , and the set  $\{\mathbf{y}_{-(50+K)}, \dots, \mathbf{y}_{-K}\}$  is discarded. Define  $\mathbf{u}_t = \Sigma_u^{1/2} \boldsymbol{\varepsilon}_t$ , where  $\Sigma_u = \sigma^2 \mathbf{I}_n$  with  $\sigma = 1$ . Another construction of  $\Sigma_u$  with nonzero off-diagonals is considered in supplementary material. Two different distributions of  $\varepsilon_{ti}$  are considered: (i) Standard normal,  $\varepsilon_{ti} \sim \text{iidN}(0, 1)$ ; (ii) Standardized mixture normal,  $\varepsilon_{ti} = (\eta_{ti} - \mu_\eta)/\sigma_\eta$  with  $\eta_{ti} = q_{ti}\xi_{ti} + (1 - q_{ti})\zeta_{ti}$ ,  $\mu_\eta = \mathbb{E}\eta_{ti}$ , and  $\sigma_\eta^2 = \mathbb{E}\eta_{ti}^2 - \mu_\eta^2$ , where  $q_{ti} \sim \text{iidBer}(\pi)$ ,  $\xi_{ti} \sim \text{iidN}(\mu_\xi, \sigma_\xi^2)$ , and  $\zeta_{ti} \sim \text{iidN}(\mu_\zeta, \sigma_\zeta^2)$ . We have chosen the parameter values  $\mu_\xi = 2$ ,  $\sigma_\xi = 2$ ,  $\mu_\zeta = 4$ ,  $\sigma_\zeta = 10$ , and  $\pi = 0.9$ , which give  $\mu_\eta = 0.4$  and  $\sigma_\eta = 3.88$ . The resulting error variable,  $\varepsilon_{ti}$ , is unimodal yet it has skewness 1.86 and kurtosis 3.53.

We focus on the model with  $K = 1$ . The coefficient matrix  $\Phi = (\phi_{ij})$  is constructed as follows. First generate  $\Psi = (\psi_{ij})$  with  $\psi_{ij} = \rho^{1+|i-j|/4} \mathbf{1}\{|i-j| \leq m\}$ . Now form an  $N \times N$  sign matrix  $\mathbf{Y}^{(r)} = (\nu_{ij}^{(r)})$  for  $r = 1, 2, \dots$ , where  $\nu_{ij}^{(r)} = 2\varphi_{ij}^{(r)} - 1$  with  $\varphi_{ij}^{(r)} \sim \text{iidBer}(0.5)$ . Repeatedly compute  $\Phi^{(r)} = \Psi \circ \mathbf{Y}^{(r)}$  for  $r = 1, 2, \dots$  until  $\lambda_{\max}(\Phi^{(r)}) \leq 0.96$ . If the process stops at the  $R$ th repetition, set  $\Phi = \Phi^{(R)}$ . We consider  $\rho = 0.4$  and  $m \in \{2, 4, 7\}$ . Observe that, except for the first and last rows, we have  $s_i = 5, 9, 15$  for  $m = 2, 4, 7$ , respectively. Set  $q = 0.1$  and consider all the combinations of  $N \in \{50, 100, 200, 300\}$  and  $T \in \{200, 300\}$ .

The model is estimated for each  $i$ th row using the R package, “glmnet.” The bootstrap procedure uses the same values of the lasso tuning parameters at the estimation stage. The CLIME estimator is constructed by the R package, “fastclime.” All the results are based on 1000 replications and 100 bootstrap samples.

### 5.2.2. Results

Table 1 summarizes the results of Procedures 1 and 2 with  $\Sigma_u = \mathbf{I}_n$  and  $\hat{m}_{ij} = \hat{\sigma}_i \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}$ . It contains three panels for  $m = 2, 4, 7$ ; The larger the value of  $m$ , the larger the number of

**Table 1.** Directional FDR and power using asymptotic and bootstrap thresholds for  $q = 0.1$ , with cross-sectionally uncorrelated errors.

| $m = 2$ (max $s_i = 5$ )  |            |      |           |      |            |      |           |      |
|---------------------------|------------|------|-----------|------|------------|------|-----------|------|
|                           | $T = 200$  |      |           |      | $T = 300$  |      |           |      |
|                           | Asymptotic |      | Bootstrap |      | Asymptotic |      | Bootstrap |      |
|                           | dFDR       | PWR  | dFDR      | PWR  | dFDR       | PWR  | dFDR      | PWR  |
| Standard normal error     |            |      |           |      |            |      |           |      |
| $N = 50$                  | 9.3        | 97.5 | 6.7       | 96.8 | 8.8        | 99.8 | 7.0       | 99.7 |
| $N = 100$                 | 10.7       | 94.7 | 6.9       | 93.2 | 9.9        | 99.4 | 7.4       | 99.2 |
| $N = 200$                 | 10.7       | 91.9 | 7.7       | 90.5 | 10.3       | 99.0 | 8.4       | 98.8 |
| $N = 300$                 | 10.4       | 89.6 | 6.8       | 87.3 | 9.7        | 98.5 | 8.5       | 98.4 |
| Mixture normal error      |            |      |           |      |            |      |           |      |
| $N = 50$                  | 10.4       | 94.2 | 5.7       | 91.9 | 9.5        | 99.0 | 6.3       | 98.7 |
| $N = 100$                 | 12.6       | 89.5 | 5.5       | 84.5 | 11.5       | 97.8 | 6.5       | 96.8 |
| $N = 200$                 | 13.1       | 87.0 | 6.6       | 82.5 | 11.7       | 97.2 | 7.6       | 96.4 |
| $N = 300$                 | 14.0       | 84.4 | 5.5       | 77.4 | 12.6       | 96.2 | 7.7       | 95.1 |
| $m = 4$ (max $s_i = 9$ )  |            |      |           |      |            |      |           |      |
|                           | $T = 200$  |      |           |      | $T = 300$  |      |           |      |
|                           | Asymptotic |      | Bootstrap |      | Asymptotic |      | Bootstrap |      |
|                           | dFDR       | PWR  | dFDR      | PWR  | dFDR       | PWR  | dFDR      | PWR  |
| Standard normal error     |            |      |           |      |            |      |           |      |
| $N = 50$                  | 8.3        | 89.1 | 6.7       | 87.9 | 7.9        | 96.7 | 6.9       | 96.4 |
| $N = 100$                 | 8.5        | 83.7 | 8.5       | 83.6 | 8.3        | 94.7 | 8.5       | 94.8 |
| $N = 200$                 | 9.3        | 78.1 | 9.6       | 78.4 | 8.8        | 91.9 | 9.6       | 92.3 |
| $N = 300$                 | 10.3       | 73.0 | 8.8       | 71.8 | 9.4        | 89.3 | 9.9       | 89.5 |
| Mixture normal error      |            |      |           |      |            |      |           |      |
| $N = 50$                  | 8.5        | 85.1 | 5.8       | 82.5 | 7.8        | 94.8 | 6.1       | 93.9 |
| $N = 100$                 | 8.9        | 79.9 | 7.2       | 78.2 | 8.7        | 92.4 | 7.6       | 91.9 |
| $N = 200$                 | 11.5       | 74.7 | 8.2       | 71.7 | 10.6       | 88.9 | 8.8       | 88.0 |
| $N = 300$                 | 12.0       | 69.9 | 6.9       | 65.3 | 11.0       | 86.1 | 8.9       | 85.1 |
| $m = 7$ (max $s_i = 15$ ) |            |      |           |      |            |      |           |      |
|                           | $T = 200$  |      |           |      | $T = 300$  |      |           |      |
|                           | Asymptotic |      | Bootstrap |      | Asymptotic |      | Bootstrap |      |
|                           | dFDR       | PWR  | dFDR      | PWR  | dFDR       | PWR  | dFDR      | PWR  |
| Standard normal error     |            |      |           |      |            |      |           |      |
| $N = 50$                  | 6.8        | 68.6 | 6.0       | 67.5 | 6.7        | 80.4 | 6.1       | 79.8 |
| $N = 100$                 | 7.6        | 60.6 | 8.0       | 61.0 | 7.6        | 74.2 | 8.0       | 74.6 |
| $N = 200$                 | 9.5        | 53.0 | 9.0       | 52.5 | 8.6        | 67.8 | 9.1       | 68.2 |
| $N = 300$                 | 9.4        | 50.1 | 9.3       | 50.1 | 8.5        | 65.3 | 10.1      | 66.4 |
| Mixture normal error      |            |      |           |      |            |      |           |      |
| $N = 50$                  | 6.4        | 65.5 | 4.9       | 63.2 | 6.1        | 77.9 | 5.1       | 76.6 |
| $N = 100$                 | 7.9        | 58.1 | 6.8       | 56.7 | 7.9        | 71.8 | 7.2       | 71.1 |
| $N = 200$                 | 9.4        | 51.1 | 7.5       | 49.3 | 9.0        | 65.6 | 8.3       | 65.0 |
| $N = 300$                 | 10.5       | 48.9 | 7.5       | 46.5 | 9.6        | 63.4 | 9.1       | 62.9 |

nonzeros in  $\Phi$ . The results with  $\Sigma_u$  with nonzero off-diagonals are qualitatively very similar; see Table F4 in supplementary material.

**Table 2.** FDR and power using e-BH thresholds for  $q = 0.1$ , with cross-sectionally uncorrelated errors.

| $m = 2 (\max s_j = 5)$  |           |      |                     |      |           |      |                     |      |
|-------------------------|-----------|------|---------------------|------|-----------|------|---------------------|------|
| $f(x)$                  | $T = 200$ |      |                     |      | $T = 300$ |      |                     |      |
|                         | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      |
|                         | FDR       | PWR  | FDR                 | PWR  | FDR       | PWR  | FDR                 | PWR  |
| Standard normal error   |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 1.8       | 93.2 | 1.3                 | 92.0 | 1.4       | 99.0 | 0.9                 | 98.7 |
| $N = 100$               | 2.2       | 88.4 | 1.5                 | 86.5 | 1.8       | 98.0 | 1.2                 | 97.5 |
| $N = 200$               | 1.7       | 82.9 | 1.2                 | 80.8 | 1.5       | 96.9 | 1.0                 | 96.3 |
| $N = 300$               | 1.4       | 78.6 | 1.1                 | 76.6 | 1.2       | 95.7 | 0.8                 | 95.0 |
| Mixture normal error    |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 2.5       | 87.7 | 1.8                 | 86.0 | 1.9       | 97.1 | 1.4                 | 96.4 |
| $N = 100$               | 3.1       | 80.6 | 2.3                 | 78.3 | 2.4       | 94.7 | 1.7                 | 93.6 |
| $N = 200$               | 2.4       | 75.5 | 1.8                 | 73.1 | 2.0       | 92.9 | 1.4                 | 91.9 |
| $N = 300$               | 2.3       | 71.3 | 1.8                 | 69.4 | 2.3       | 71.3 | 1.5                 | 90.1 |
| $m = 4 (\max s_j = 9)$  |           |      |                     |      |           |      |                     |      |
| $f(x)$                  | $T = 200$ |      |                     |      | $T = 300$ |      |                     |      |
|                         | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      |
|                         | FDR       | PWR  | FDR                 | PWR  | FDR       | PWR  | FDR                 | PWR  |
| Standard normal error   |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 1.7       | 79.5 | 1.2                 | 77.4 | 1.4       | 91.8 | 1.0                 | 90.7 |
| $N = 100$               | 1.6       | 72.4 | 1.1                 | 69.6 | 1.5       | 88.7 | 0.9                 | 87.2 |
| $N = 200$               | 1.7       | 66.5 | 1.2                 | 63.9 | 1.5       | 85.1 | 1.0                 | 83.4 |
| $N = 300$               | 1.8       | 60.3 | 1.3                 | 58.1 | 1.5       | 81.4 | 1.0                 | 79.6 |
| Mixture normal error    |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 1.9       | 74.2 | 1.4                 | 71.9 | 1.5       | 88.4 | 1.1                 | 86.9 |
| $N = 100$               | 1.8       | 67.5 | 1.3                 | 64.5 | 1.7       | 85.0 | 1.1                 | 83.1 |
| $N = 200$               | 2.1       | 62.5 | 1.5                 | 59.9 | 1.9       | 81.0 | 1.3                 | 79.0 |
| $N = 300$               | 2.2       | 56.6 | 1.6                 | 54.4 | 1.9       | 77.1 | 1.3                 | 75.1 |
| $m = 7 (\max s_j = 15)$ |           |      |                     |      |           |      |                     |      |
| $f(x)$                  | $T = 200$ |      |                     |      | $T = 300$ |      |                     |      |
|                         | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      | $ x ^p$   |      | $\exp(c x ^\alpha)$ |      |
|                         | FDR       | PWR  | FDR                 | PWR  | FDR       | PWR  | FDR                 | PWR  |
| Standard normal error   |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 1.3       | 55.8 | 0.9                 | 53.7 | 1.1       | 69.2 | 0.8                 | 67.3 |
| $N = 100$               | 1.5       | 48.7 | 1.0                 | 46.5 | 1.4       | 63.7 | 0.9                 | 61.5 |
| $N = 200$               | 1.6       | 42.3 | 1.1                 | 40.2 | 1.4       | 58.0 | 0.9                 | 55.9 |
| $N = 300$               | 1.7       | 39.8 | 1.2                 | 38.0 | 1.4       | 55.7 | 1.0                 | 53.9 |
| Mixture normal error    |           |      |                     |      |           |      |                     |      |
| $N = 50$                | 1.3       | 52.0 | 0.9                 | 49.6 | 1.1       | 65.9 | 0.8                 | 63.8 |
| $N = 100$               | 1.6       | 45.9 | 1.1                 | 43.5 | 1.5       | 60.7 | 1.1                 | 58.4 |
| $N = 200$               | 1.9       | 40.2 | 1.3                 | 38.1 | 1.7       | 55.2 | 1.1                 | 53.0 |
| $N = 300$               | 1.9       | 38.1 | 1.4                 | 36.4 | 1.7       | 53.2 | 1.2                 | 51.3 |

As can be seen, both procedures control the dFDR to be around the predetermined level  $q = 0.1$ , maintaining the high power. In particular, the good performance for large  $N \geq T$  is very encouraging. The asymptotic threshold tends to produce slightly larger dFDRs than  $q$  while the bootstrap one is more conservative. The mixture normal error moderately amplifies the tendency. Despite this conservativeness, the power based on the bootstrap threshold is almost identical to that based on the asymptotic one. The exaggeration and the conservativeness by both procedures are mitigated if  $T$  is getting large. As expected, when the nonzero elements in  $\Phi$  increase (i.e.,  $m$  rises), the power goes down. However, it quickly rises as  $T$  increases.

The results of Procedure 3 based on the transformed  $t$ -statistic,  $f(T_{ij})$ , is reported in Table 2. Two transformation functions  $f(x)$  are considered:  $|x|^p$  with  $p = 10$  and  $\exp(c|x|^\alpha)$  with  $c = 3$  and  $\alpha = 1$ . Both give very similar FDR and power. Their FDR are more conservative than the asymptotic

and bootstrap FDR, but the associated reduction in power is relatively small. In conclusion, the e-BH procedure can be a reliable and robust alternative.

## 6. Two Empirical Applications

We apply our proposed methods to two large datasets to discover the underlying Granger-causal networks. Section 6.1 investigates the large macroeconomic and financial variables. Section 6.2 analyzes the regional house price growths in the United Kingdom. All the results in this section are based on  $K = 1$  and

$$\hat{m}_{ij} = \hat{\sigma}_i \sqrt{\hat{\omega}_j' \hat{\Sigma}_x \hat{\omega}_j}.$$

### 6.1. Large Macroeconomic Variables

The FRED-MD data file of May 2019 is obtained from McCracken's website, and the variables are transformed to be stationary as instructed by McCracken and Ng (2016). The data consists of a balanced panel of 128 monthly series spanning the period from June 1999 to May 2019. All series are standardized before the analysis. Following McCracken and Ng (2016), the series are categorized into eight groups: **G1**, Output and Income; **G2**, Labour Market; **G3**, Consumption, Orders and Inventories; **G4**, Housing; **G5**, Interest and Exchange Rate; **G6**, Prices; **G7**, Money and Credit; **G8**, Stock Market. (The group order is different from McCracken and Ng (2016).) The variables are numbered from 1 to 128, and the descriptions for all the variables are reported in Table G1 in supplementary material.

We estimate a VAR(1) model. The asymptotic and bootstrap thresholds for the  $t$ -ratios with  $q = 0.05$  were 2.88 and 4.41, respectively. Here we summarize the result with Procedure 2 in Figure 4 as a network Granger causality diagram. The result with Procedure 1 is reported in supplementary material. The nodes represent the variables, and their colors show the eight categories. The size of a node indicates the number of variables it significantly predicts; the larger the node size, the more variables it can predict. The arrows show the direction of the Granger causality. The self-lag effects are excluded in the figure. The main dynamic inter-relations are clustered within the groups, yet interesting interlinkages between the variable groups are observed. In particular, Price variables are clustered together, and 11 price variables are caused by variable 112, *Real M2 Money Supply*. Price variable 94, *Crude Oil*, also Granger-causes seven other price variables. Finally, *Real Manufactures and Trade Industries Sales* (variable 49) Granger-cause three producer price indices (variables 90, 91, 92). These findings make a lot of sense from an economic point of view. In addition, it is easy to identify the variables that cause many other variables (many edges come out from the node) and those which are caused by many other variables (the node surrounded by many pointing arrows); see the *Housing* variable cluster, for example.

### 6.2. UK Regional House Price Growths

We obtained the monthly average house prices at the local authority district level, published in November 2021 by HM Land Registry in the United Kingdom. Before analysis, we seasonally adjusted the prices, then deflated them by the U.K.

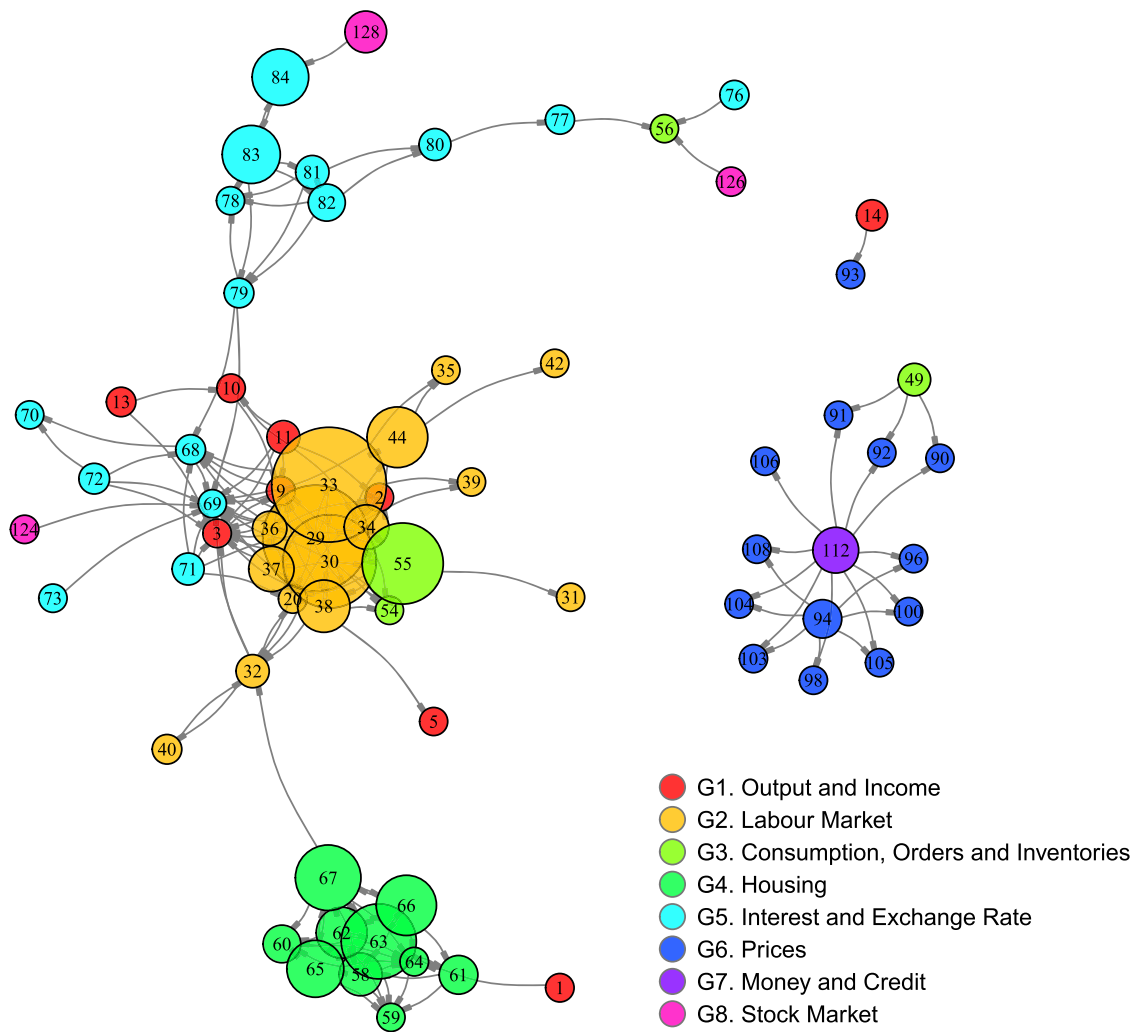


Figure 4. Network Granger-causality: 128 macroeconomic variables, bootstrap  $t_0$ ,  $q = 0.05$ .

consumer price index (CPI).<sup>1</sup> Denoting by  $HP_{it}$  the seasonally adjusted real house price of the district  $i$  at month  $t$ , the monthly house price growth is computed as  $\Delta hp_{it} = \log(HP_{it}/HP_{it-1})$ . In this analysis, we choose variables of 86 districts of Scotland, Wales, and the London area, spanning 209 months, from February 2004 to June 2021. The variables are numbered from 1 to 86, and the full list of the district names is reported in Table G2 in supplementary materials. All the series are demeaned before the analysis.

We estimate a VAR(1) model. The asymptotic and bootstrap thresholds for the  $t$ -ratios with  $q = 0.05$  were 2.73 and 4.22, respectively. Following the previous subsection, we summarize the result with the bootstrap threshold in Figure 5 as a network Granger causality diagram. The result with the asymptotic threshold is reported in supplementary material.

The results show that house price growth causality networks are more or less clustered in each of the three regions. However, there are interesting inter-regional Granger-causal relationships. The London network Granger-causes variables 21 and 24, which are the house price growth in the City of Edinburgh and West

Lothian. These two regions comprise a large part of Edinburgh, the capital of Scotland. Furthermore, the London regional network is shown to Granger-cause and also to be Granger-caused by Cardiff, the capital of Wales, variable 78. These results suggest that house price fluctuations in London interact dynamically with those in the capitals of other countries in the United Kingdom.

There is one unique inter-regional Granger Causality. Falkirk in Scotland (Variable 6) has a significant concentration of arrows from London, Scotland, and Wales. A little searching leads to the BBC News of July 21, 2020 reporting that the Scottish and UK governments have pledged 90 million pounds in “Growth Deal” funds to stimulate the economy around Falkirk.<sup>2</sup> The deal was signed off by the United Kingdom, Scottish governments, and Falkirk council on December 21, 2021.<sup>3</sup> This large deal may have attracted investment to the Falkirk property market and was statistically identified as a Granger Causality for house price fluctuations in Falkirk.

<sup>1</sup>The CPI index (D7BT, not seasonally adjusted) is obtained from Office for National Statistics, UK. The house prices and the CPI are seasonally adjusted using the R package “seasonal.”

<sup>2</sup><https://www.bbc.co.uk/news/uk-scotland-tayside-central-53471904>

<sup>3</sup><https://www.bbc.co.uk/news/uk-scotland-scotland-business-59734937>

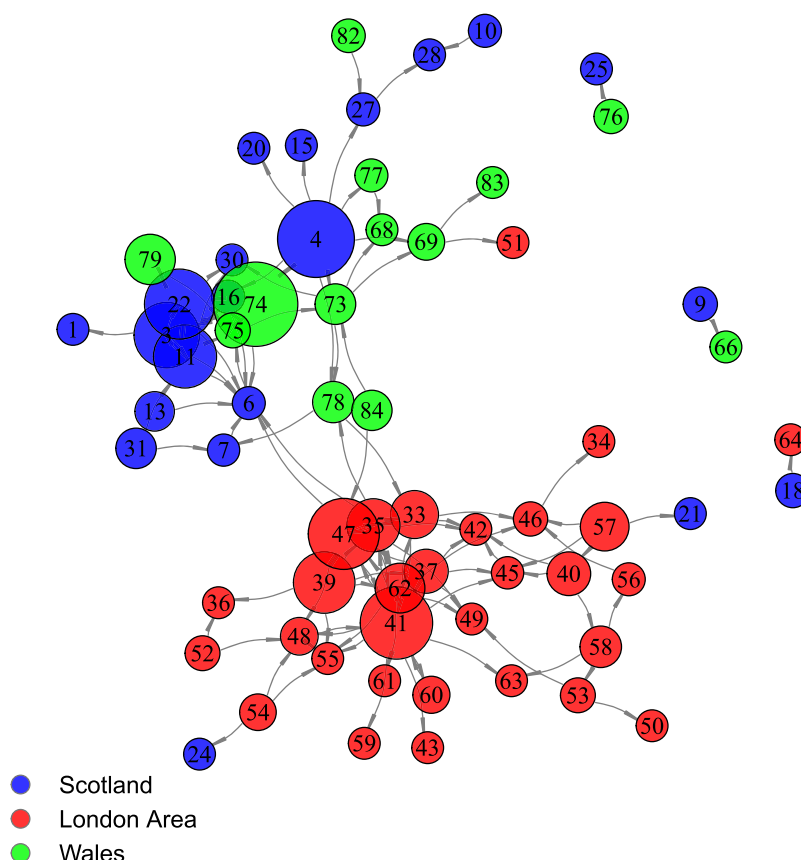


Figure 5. Network Granger causality: 86 UK regional house prices, bootstrap  $t_0$ ,  $q = 0.05$ .

## 7. Conclusion

This article has proposed multiple testing procedures that control the FDR for discovering the network Granger causality in high-dimensional VAR models. The validity of our inference-based framework is supported by the theory, simulation studies, and two empirical applications. We hope that the methods will enable us to stably discover networks inherent in various high-dimensional time series, serving as a clue for new theories in their domains.

We can consider some directions for future research: (i) Eichler (2007) investigates a *path diagram* that is composed of the Granger-causal network and contemporaneous connections in the error covariance matrix. As considered in Barigozzi and Brownlees (2019), extending our methods to include detection of the contemporaneous correlation networks is interesting and important; (ii) We have not included a serious discussion of lag order selection. Stokes and Purdon (2017) argue that both too large and too small lag order selections will lead to spurious Granger causality findings. Apparently, investigating the “optimal” choice of  $K$ , such as combining the regularization-based lag selection method of Nicholson et al. (2020) with our framework, can be another important topic.

## Supplementary Materials

Section A: Proof of Theorems, Section B: Proof of Propositions, Section C: Lemmas and their proofs, Section D: Precision Matrix Estimation, Section E: Additional Results on Robustification, Section F: Additional Experimen-

tal Results, Section G: List of Variable Names in Empirical Applications, Section H: Additional Results of Empirical Applications.

## Acknowledgments

The authors thank the editor, associate editor, and anonymous referees for their valuable comments and suggestions, which significantly improved the article.

## Disclosure Statement

The authors report there are no competing interests to declare.

## Funding

This work was supported by JSPS KAKENHI Grant Numbers 19K13665, 20H01484, 20H05631, 21H00700, and 21H04397.

## References

- Babii, A., Ghysels, E., and Striaukas, J. (2021), “High-Dimensional Granger Causality Tests with an Application to vix and News,” SSRN: <https://ssrn.com/abstract=3615718>. [2386]
- Barigozzi, M., and Brownlees, C. (2019), “NETS: Network Estimation for Time Series,” *Journal of Applied Econometrics*, 34, 347–364. [2386,2395]
- Basu, S., Li, X., and Michailidis, G. (2019), “Low Rank and Structured Modeling of High-Dimensional Vector Autoregressions,” *IEEE Transactions on Signal Processing*, 67, 1207–1222. [2386,2387]
- Basu, S., and Michailidis, G. (2015), “Regularized Estimation in Sparse High-Dimensional Time Series Models,” *Annals of Statistics*, 43, 1535–1567. [2386]



- Basu, S., Shojai, A., and Michailidis, G. (2015), "Network Granger Causality with Inherent Grouping Structure," *Journal of Machine Learning Research*, 16, 417–453. [2385,2387]
- Benjamini, Y., and Hochberg, Y. (1995), "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society, Series B*, 57, 289–300. [2386,2389]
- Bonferroni, C. E. (1935), "Il calcolo delle assicurazioni su gruppi di teste," *Studi in Onore del Professore Salvatore Ortu Carboni*, 13–60. [2386]
- Brüggemann, R., Jentsch, C., and Trenkler, C. (2016), "Inference in Vars with Conditional Heteroskedasticity of Unknown Form," *Journal of Econometrics*, 191, 69–85. [2388]
- Cai, T., Liu, W., and Luo, X. (2011), "A Constrained  $\ell_1$  Minimization Approach to Sparse Precision Matrix Estimation," *Journal of the American Statistical Association*, 106, 594–607. [2388,2390]
- Candès, E., and Tao, T. (2007), "The Dantzig Selector: Statistical Estimation When  $p$  is Much Larger than  $n$ ," *Annals of Statistics*, 35, 2313–2351. [2386]
- Candès, E. J., Fan, Y., Janson, L., and Lv, J. (2018), "Panning for Gold: 'model-x' Knockoffs for High Dimensional Controlled Variable Selection," *Journal of the Royal Statistical Society, Series B*, 80, 551–577. [2386]
- Chi, C.-M., Fan, Y., Ing, C.-K., and Lv, J. (2021), "High-Dimensional Knockoffs Inference for Time Series Data," arXiv:2112.09851. [2386]
- Davidson, R., and Flachaire, E. (2008), "The Wild Bootstrap, Tamed at Last," *Journal of Econometrics*, 146, 162–169. [2389]
- Davis, R. A., Zang, P., and Zheng, T. (2016), "Sparse Vector Autoregressive Modeling," *Journal of Computational and Graphical Statistics*, 25, 1077–1096. [2386]
- Dufour, J.-M., and Renault, E. (1998), "Short Run and Long Run Causality in Time Series: Theory," *Econometrica*, 66, 1099–1125. [2385]
- Eichler, M. (2007), "Granger Causality and Path Diagrams for Multivariate Time Series," *Journal of Econometrics*, 137, 334–353. [2385,2395]
- (2012a), "Causal Inference in Time Series Analysis," in *Causality: Statistical Perspectives and Applications*, eds. C. Berzuini, P. Dawid, and L. Bernardinelli, pp. 327–354, Chichester: Wiley. [2385]
- (2012b), "Graphical Modelling of Multivariate Time Series," *Probability Theory and Related Fields*, 153, 233–268. [2385]
- Fujita, A., Sato, J. R., Garay-Malpartida, H. M., Yamaguchi, R., Miyano, S., Sogayar, M. C., and Ferreira, C. E. (2007), "Modeling Gene Expression Regulatory Networks with the Sparse Vector Autoregressive Model," *BMC Systems Biology*, 1, 1–11. [2386]
- Gonçalves, S., and Kilian, L. (2004), "Bootstrapping Autoregressions with Conditional Heteroskedasticity of Unknown Form," *Journal of Econometrics*, 123, 89–120. [2388]
- Granger, C. W. J. (1969), "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods," *Econometrica*, 37, 424–438. [2385,2386]
- Guðmundsson, G. S., and Brownlee, C. (2021), "Detecting Groups in Large Vector Autoregressions," *Journal of Econometrics*, forthcoming. [2387]
- Hafner, C. M., and H. Herwartz (2009), "Testing for Linear Vector Autoregressive Dynamics Under Multivariate Generalized Autoregressive Heteroskedasticity," *Statistica Neerlandica*, 63, 294–323. [2388]
- Han, F., Lu, H., and Liu, H. (2015), "A Direct Estimation of High Dimensional Stationary Vector Autoregressions," *Journal of Machine Learning Research*, 16, 3115–3150. [2386]
- Holm, S. (1979), "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, 6, 65–70. [2386]
- Hosoya, Y. (1977), "On the Granger Condition for Non-causality," *Econometrica*, 45, 1735–1736. [2385]
- Javanmard, A., and Javadi, H. (2019), "False Discovery Rate Control via Debiased Lasso," *Electronic Journal of Statistics*, 13, 1212–1253. [2388]
- Javanmard, A., and Montanari, A. (2014), "Confidence Intervals and Hypothesis Testing for High-Dimensional Regression," *Journal of Machine Learning Research*, 15, 2869–2909. [2387]
- Jing, B.-Y., Shao, Q.-M., and Wang, Q. (2003), "Self-Normalized Cramér-type Large Deviations for Independent Random Variables," *Annals of Probability*, 31, 2167–2215. [2391]
- Kilian, L. (1999), "Finite-Sample Properties of Percentile and Percentile-Bootstrap Confidence Intervals for Impulse Responses," *Review of Economics and Statistics*, 81, 652–660. [2388]
- Kock, A. B., and Callot, L. (2015), "Oracle Inequalities for High Dimensional Vector Autoregressions," *Journal of Econometrics*, 186, 325–344. [2386]
- Lin, J., and Michailidis, G. (2017), "Regularized Estimation and Testing for High-Dimensional Multi-Block Vector-Autoregressive Models," *Journal of Machine Learning Research*, 18, 1–49. [2387]
- Liu, W. (2013), "Gaussian Graphical Model Estimation with False Discovery Rate Control," *Annals of Statistics*, 41, 2948–2978. [2388]
- Lozano, A. C., Abe, N., Liu, Y., and Rosset, S. (2009), "Grouped Graphical Granger Modeling for Gene Expression Regulatory Networks Discovery," *Bioinformatics*, 25, i110–i118. [2386]
- Lütkepohl, H. (2005), *New Introduction to Multiple Time Series Analysis*, Berlin, Heidelberg: Springer. [2385]
- Mammen, E. (1993), "Bootstrap and Wild Bootstrap for High Dimensional Linear Models," *Annals of Statistics*, 21, 255–285. [2389]
- McCracken, M. W., and Ng, S. (2016), "Fred-md: A Monthly Database for Macroeconomic Research," *Journal of Business & Economic Statistics*, 34, 574–589. [2393]
- Nicholson, W. B., Wilms, I., Bien, J., and Matteson, D. S. (2020), "Testing High Dimensional Forecasting via Interpretable Vector Autoregression," *Journal of Machine Learning Research*, 21, 1–52. [2387,2395]
- Reid, S., Tibshirani, R., and Friedman, J. (2016), "A Study of Error Variance Estimation in Lasso Regression," *Statistica Sinica*, 26, 35–67. [2388]
- Shojai, A., and Fox, E. B. (2021), "Granger Causality: A Review and Recent Advances," arXiv:2105.02675. [2385]
- Shu, H., and Nan, B. (2019), "Estimation of Large Covariance and Precision Matrices from Temporally Dependent Observations," *Annals of Statistics*, 47, 1321–1350. [2390]
- Sims, C. A. (1972), "Money, Income, and Causality," *American Economic Review*, 62, 540–552. [2385]
- Stokes, P. A., and Purdon, P. L. (2017), "A Study of Problems Encountered in Granger Causality Analysis from a Neuroscience Perspective," *Proceedings of the National Academy of Sciences*, 114, E7063–E7072. [2395]
- Tibshirani, R. (1996), "Regression Shrinkage and Selection via the Lasso," *Journal of the Royal Statistical Society, Series B*, 58, 267–288. [2386]
- Uematsu, Y., and Yamagata, T. (2023), "Inference in Sparsity-Induced Weak Factor Models," *Journal of Business & Economic Statistics*, 41, 126–139. [2388]
- van de Geer, S., Bühlmann, P., Ritov, Y., and Dezeure, R. (2014), "On Asymptotically Optimal Confidence Regions and Tests for High-Dimensional Models," *Annals of Statistics*, 42, 1166–1202. [2387]
- Vovk, V., and Wang, R. (2021), "E-values: Calibration, Combination and Applications," *The Annals of Statistics*, 49, 1736–1754. [2389]
- Wang, R., and Ramdas, A. (2022), "False Discovery Rate Control with E-values," *Journal of the Royal Statistical Society, Series B*, 84, 822–852. [2389]
- Zhang, C.-H., and Zhang, S. S. (2014), "Confidence Intervals for Low Dimensional Parameters in High Dimensional Linear Models," *Journal of the Royal Statistical Society, Series B*, 76, 217–242. [2387]
- Zheng, L., and Raskutti, G. (2019), "Testing for High-Dimensional Network Parameters in Auto-Regressive Models," *Electronic Journal of Statistics*, 13, 4977–5043. [2386]
- Zhu, K., and Liu, H. (2020), "Confidence Intervals for Parameters in High-Dimensional Sparse Vector Autoregression," arXiv:2009.09462. [2386]